Lecture 4: Nonlinear Forecasting Methods

Ivy Yang[1]

[1]Economics & Finance, Business School, Southern Illinois University - Edwardsville

Why linear models can miss business reality

Definition: Nonlinearity

A nonlinear forecasting relationship means that the effect of one predictor does not stay constant across its whole range.

Linear models are a useful default, but many financial relationships are not linear. A small VIX increase during a calm period may have only a modest effect on equity returns, but a sharp volatility spike during a stress episode can produce a disproportionately large negative return that a linear model systematically misses. The main tradeoff is simple interpretation versus realistic shape. For beginner forecasters, the practical rule is to start linear, diagnose the failure pattern, and add flexibility only when the data and the business story point in the same direction.

Example 4.1. *In the notebook, same-day VIX changes are plotted against S&P 500 returns using data from 2015–2024. A linear fit through this scatter underestimates the severity of large negative moves during volatility spikes, while a quadratic model captures the accelerating downside curvature more accurately. Points are color-coded by VIX regime (calm, normal, and stressed) to show how the nonlinear relationship concentrates in stress episodes.*

Polynomial terms and interactions

Definition: Polynomial term

A polynomial term adds curvature by including powers such as x^2 or x^3 in the regression.

Definition: Interaction term

An interaction term allows the effect of one predictor to depend on the level of another predictor.

Polynomial terms are the easiest extension of a linear model because the equation remains linear in coefficients even though the predictor shape bends. Interactions are equally important because real forecasting effects often depend on context. The predictive content of lagged stock returns may differ substantially depending on whether the market is in a calm or stressed volatility state.

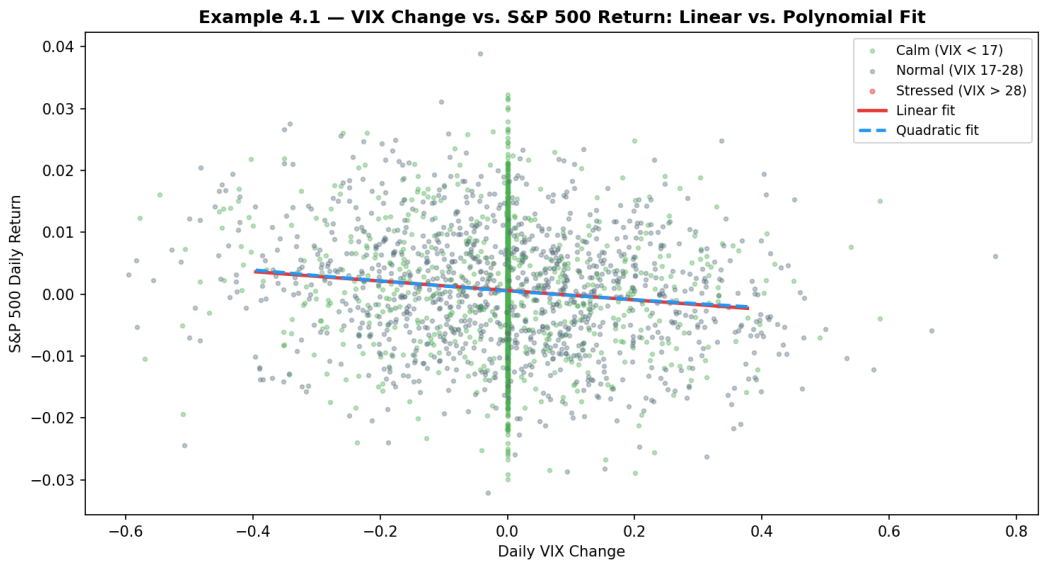

Figure 1: Example 4.1 notebook output: linear fit versus quadratic VIX-return curve, color-coded by volatility regime.

The tradeoff is that added flexibility improves fit only if it reflects a real mechanism rather than accidental noise.

$$r_t^{SP500} = \beta_0 + \beta_1 \Delta\text{VIX}_t + \beta_2 \Delta\text{VIX}_t^2 + \beta_3(\mathbf{1}_{\text{HighVIX},t} \times r_{t-1}^{MSFT}) + u_t$$

Example 4.2. *The notebook compares a model using only the lagged MSFT return against a model that adds a VIX regime interaction (VIX $\geq$ 25). The interaction model captures the empirical pattern that MSFT's predictive content for S&P 500 returns is amplified in stressed markets, where correlation across assets tends to rise sharply.*

Decision trees as forecasting rules

Definition: Decision tree

A decision tree forecasts by splitting the data into branches using simple if-then rules, such as "if VIX is high" or "if yesterday's return is negative".

Decision trees are appealing for undergraduates because they turn forecasting logic into a sequence of business rules. Instead of forcing one global slope, a tree can allow one rule for days when yesterday's S&P 500 return was positive, another for negative-return days, and a further split based on whether VIX is rising or falling. The gain is interpretability of local regimes. The cost is that a single tree can become unstable: a small data change may produce a different split and a different story.

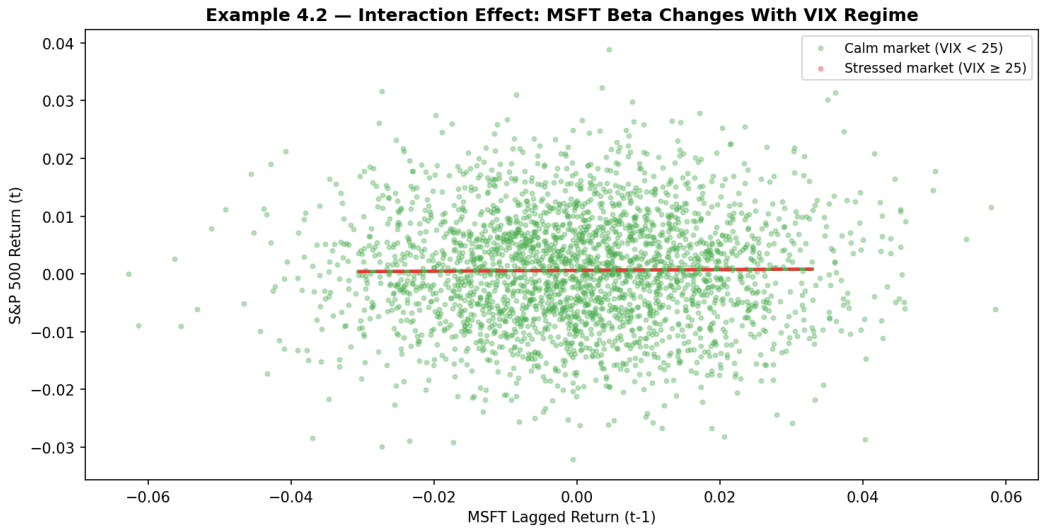

Figure 2: Example 4.2 notebook output: lagged MSFT beta differs between calm and stressed VIX regimes.

Example 4.3. *In the notebook, a shallow decision tree (max depth 3) is fitted on lagged S&P 500 returns and daily VIX changes. The resulting rectangular partitions in the two-dimensional feature space show how a tree creates regime-specific forecast rules: calm-and-up, calm-and-down, stressed-and-up, and stressed-and-down market states each receive a different predicted return.*

Model flexibility and overfitting in trees

Definition: Tree depth

Tree depth is the number of split layers in a decision tree. Greater depth means more flexibility and higher overfitting risk.

Flexible models are powerful precisely because they can follow subtle patterns, but that same flexibility makes them more vulnerable to memorizing accidental variation. In forecasting, this is especially dangerous because the future never repeats the training sample exactly. The relevant tradeoff is not "most flexible" versus "least flexible" but "best validation performance" versus "false confidence from excessive complexity".

Example 4.4. *The notebook increases tree depth from 1 to 13 on the S&P 500 return forecasting task. Training RMSE keeps falling as the tree grows, but validation RMSE starts rising after a moderate depth. This is the same Lecture 2 logic applied in a nonlinear setting: the best model is chosen by validation performance, not in-sample fit.*

Tree hyperparameters are regularization. Tree depth, minimum samples per leaf, and maximum features at each split are all forms of regularization. Just as λ in ridge and lasso constrains coeffi-

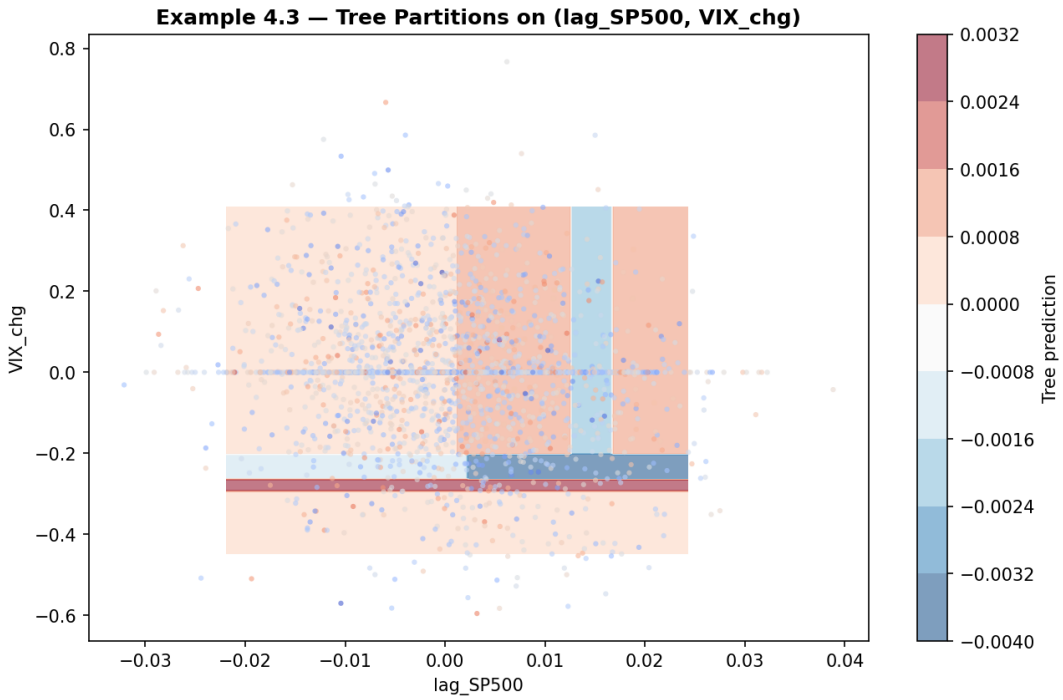

Figure 3: Example 4.3 notebook output: tree partitions on (lag_SP500, VIX_chg) create regime-specific forecast rules.

cient size, these parameters constrain tree complexity. The bias–variance trade-off from Lecture 3 applies directly: more restrictive settings add bias but reduce variance and improve out-of-sample stability.

Regularization tool	**In linear models (L3)**	**In trees (L4)**
Constraint on complexity	λ penalty on β size	`max_depth`, `min_samples_leaf`
Tuning method	Validation RMSE grid search	Validation RMSE grid search
Effect	Shrinks coefficients	Limits split depth

Random forests as averaged trees

Definition: Random forest

A random forest combines many decision trees and averages their predictions to reduce variance and improve stability.

Random forests are useful because they keep the nonlinear, interaction-friendly structure of trees while reducing the instability of any one tree. The idea is simple: many noisy trees vote together, and the average is usually less erratic than a single tree. The main tradeoff is interpretability versus robustness. A single tree is easy to explain; a forest usually forecasts better but is harder to summarize in one sentence.

Figure 4: Example 4.4 notebook output: tree depth lowers train error faster than validation error.

Example 4.5. *The notebook compares the best single decision tree with a random forest (200 trees) on the S&P 500 return forecasting task. The forest produces lower test-set RMSE and a smoother forecast path over the 80-day test window because averaging across trees dampens the noise from individual splits.*

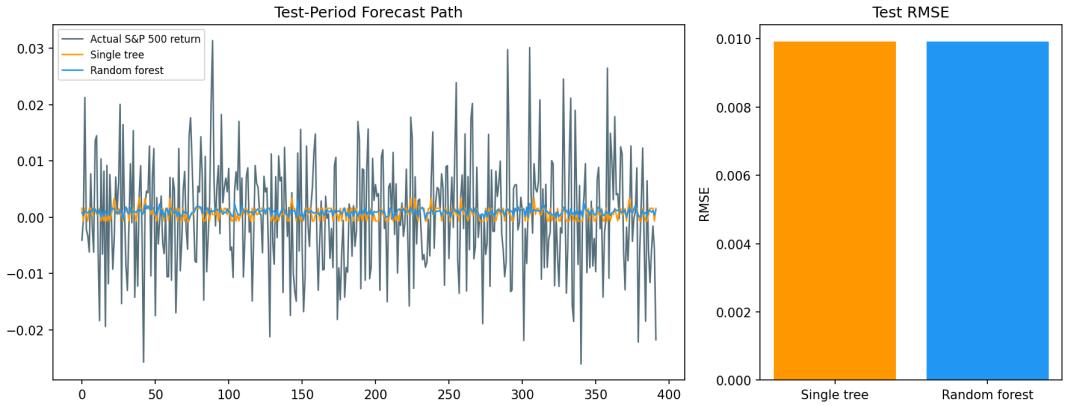

Figure 5: Example 4.5 notebook output: random forest improves stability relative to a single tree on S&P 500 returns.

Feature importance in random forests

Definition: Feature importance

Feature importance measures how much each predictor reduces forecast error across all trees in the forest, expressed as a fraction of the total reduction.

One practical advantage of random forests is automatic variable ranking. In `sklearn`, the `feature_importan` attribute assigns each predictor a score between 0 and 1 that sums to 1 across all features.

- **High importance**: the predictor consistently helps reduce prediction error across many trees and splits.
- **Low importance**: the predictor rarely improves splits and can often be dropped without hurting validation performance.

Example 4.6. *In the notebook, feature importance scores for the S&P 500 return model highlight a small set of dominant predictors, with lagged individual stock returns (MSFT, AAPL, JPM, XOM) contributing at varying levels across runs. This ranking tells the analyst which signals most deserve monitoring and data-quality investment. A companion chart compares the rolling prediction variance of the single tree versus the forest, showing that the forest's averaging substantially narrows day-to-day forecast instability.*

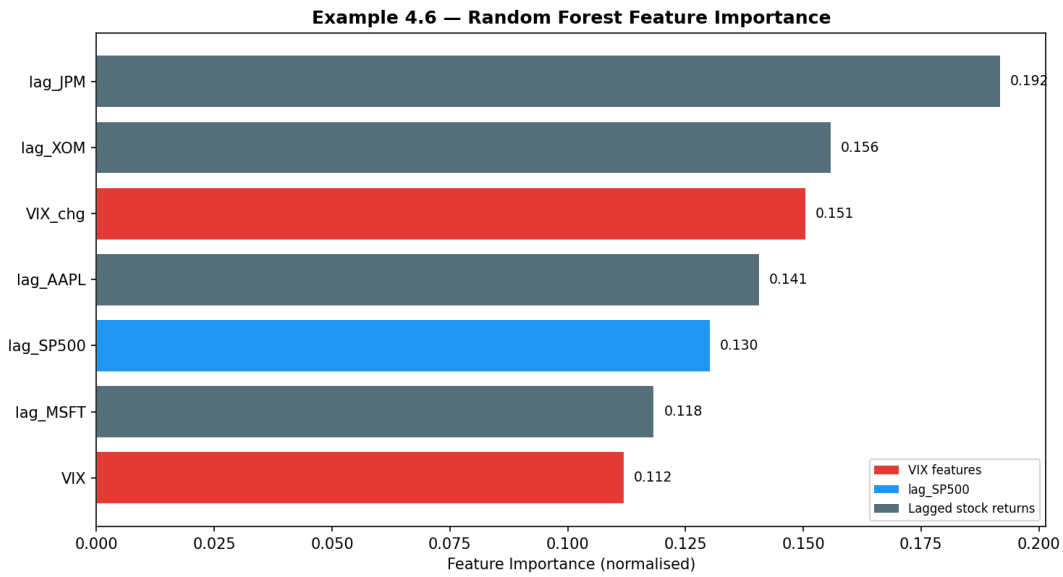

Figure 6: Example 4.6 notebook output: random forest feature importance for the S&P 500 return model.

Caution. Feature importance scores describe the model, not the underlying causal structure. Two highly correlated predictors may split importance between them—similar to the multicollinearity issue in ridge and lasso (Lecture 3). Use importance rankings for prioritization, not for causal attribution.

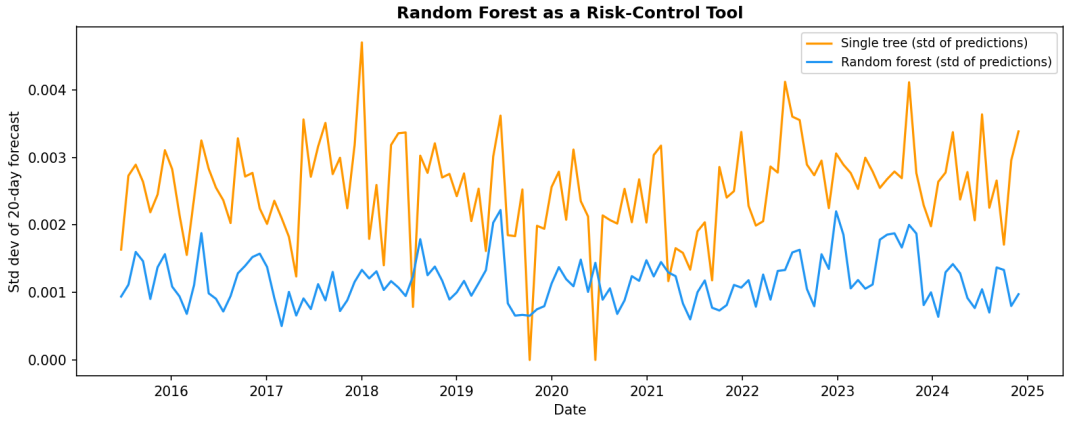

Figure 7: Example 4.6 companion: rolling prediction variance – single tree versus random forest.

Random forest as a risk-control tool

Definition: Variance reduction

Averaging predictions from many independent trees reduces forecast variance without requiring assumptions about the functional form of the relationship.

Random forest reduces the instability of individual trees by aggregating their predictions. This is why institutions—banks, asset managers, and central banks—often favor ensemble methods over single models when stakes are high:

- A single decision tree can change its forecast substantially after one unusual observation.
- A random forest averages that noise away, producing more stable forecasts over time.
- Stability makes the model easier to defend in committees and easier to audit.

Key trade-off.

	Single decision tree	**Random forest**
Interpretability	High: readable if-then rules	Lower: average of many rules
Variance	High: sensitive to small data changes	Lower: averaging stabilizes output
Out-of-sample RMSE	Often weaker	Usually stronger
Institutional preference	Limited	Preferred for risk-sensitive decisions

Forecasting rules for nonlinear methods

Definition: Flexibility discipline

Use nonlinear models only when validation, timing discipline, and business interpretation all support the added complexity.

The forecasting lesson is not "trees beat regression" or "machine learning is always better." The correct lesson is that more flexible models should only be trusted when they obey the same time-order rules as linear models, beat simple benchmarks out of sample, and produce behavior that a decision-maker can still understand. In business settings, a slightly less accurate model is often preferable if its forecast logic is easier to monitor and defend.

Example 4.7. *In the notebook conclusion, the preferred nonlinear model is chosen using a chronological train/validation/test split rather than in-sample fit. This keeps Lecture 4 connected to the discipline introduced in Lectures 2 and 3.*

Interview Questions This Lecture Prepares You To Answer

- **"Why did you choose random forest?"**
  "Random forest reduces the instability of a single decision tree by averaging many trees fitted on different data subsets and feature subsets. It handles nonlinear effects and interactions automatically without requiring manual feature engineering, and it is more stable than any single tree."

- **"What does random forest buy you over linear models?"**
  "Linear models assume the effect of a predictor stays the same across its full range. Random forest allows the effect to vary depending on the values of other variables, which is more realistic when relationships have thresholds or regime-dependent behavior—for example, lagged stock returns whose predictive content is amplified during high-volatility market stress episodes."

- **"What are the limitations of random forest?"**
  "Three main limitations. First, it is harder to explain than a single tree or a linear model: there is no one-sentence summary of the model logic. Second, it does not extrapolate well beyond the range of training data—the forest can only predict within the distribution it has seen. Third, like all models, it must be validated out of sample and monitored after deployment; strong validation performance does not guarantee future stability."